err.Error() to HTTP response enumeration (PX-F18, §11.4.276(C) class enumeration), rework 3 (2026-11-13)

Field	Value
Date	2026-11-13
Roadmap item	PX-F18 (CAT-PREEXIST-3)
Review base	main @ 9ed6577b (the base the NO-GO review of 2026-11-12 read)
Current base	main @ 713a4f380 (every "HEAD" below)
Scope	every non-test .go file under catalog-api (all packages, `main.go` included) plus the other Go modules of the repository that serve HTTP (OCU-CUDA-Sidecar; `scripts/test-infra/nas_clients` scanned: 0 hits)
Status	enumeration only; no source file changed. Replaces the doc the review rejected (it was never landed)
Verdict labels	PROVISIONAL, review owed. Nothing here is a review GO

Why this is a rework

The NO-GO review (docs/FOLLOW_UPS_NEXT_PHASE.md, "PX-F18 enumeration doc review: NO-GO") found one unlisted class and five defects of the instrument and the wording. Each is closed in this document as follows (evidence is in the sections named).

Review finding	Closure	Where
F1: `utils.SendErrorResponse(c,status,msg,err)`	The scanner now matches a bare `err` argument. At the re-	"Class and th

Review finding	Closure	Where
puts `err.Error()` into `details`; 109 call sites in 11 files (media_entity 35, collection 15, comic_pages 14, stats 10, media 10, search 6, recommendation 6, challenge 5, browse 5, media_browse 2, middleware/auth.go 1; auth.go:167 sends JWT parse text on 401)	view base it reproduces the review's numbers exactly: 109 bare-err call sites in the same 11 files with the same per-file counts, plus one more SendErrorResponse site that passes `err.Error()` as the *message* (collection_handler.go:335). The class is enumerated per site (rows B1..B226) and the mechanism is verified: base `utils/response.go:31` has `response.Details = err.Error()`; HEAD has no assignment to `Details` anywhere in `utils/`	per-sit table
F2: internal/metrics/health.go:157 (unauthenticated /health/deep leaked the DB ping error); monitoring.go:251 (unwired)	Both are rows H1, H2. They are struct/map-literal flows that a sink-call scanner cannot see, so a second instrument (every zero-argument `.Error()` call, any context) was added. health.go:157 is closed at HEAD (fixed text, error logged). monitoring.go:251 is still present at HEAD and is unwired (no non-test importer of `catalogizer/internal/monitoring`)	"Class 2", row H1/H2, "HEAD resid-ual" R1
F3: auth rows should be LEAK	Reclassified. At the review base `AuthService.Login` returns `fmt.Errorf("failed to get user: %w", err)` and similar chains (auth_service.go:105,147,170,177,183,189) and the handler sends `err.Error()` on 401, so the rows are LEAK, with authored sentinels (locked, disabled, pending) mixed in. The 20 former UNKNOWN rows were each resolved by reading the	per-sit table, colum Class

Review finding	Closure	Where
	producer (rule list in "Classification rules")	
F4: middleware rows are latent	Reclassified LATENT (rows with `internal/middleware/middleware.go`). ErrorHandler is wired (`main.go:1038` at base, `:1045` at HEAD), but its two leaking branches need gin `ErrorTypeBind` or `ErrorTypePublic` errors, and no non-test code produces either (grep for `.Bind(`, `BindJSON`, `MustBindWith`, `AbortWithError(`, `ErrorTypePublic`: 0 hits outside the middleware file; `ShouldBind*` does not append to `c.Errors`). Third-party middleware that might append such errors was not enumerated: UNCONFIRMED	per-sit table
F5: limits text wrong	The old text said "no such helper found by name search"; SendErrorResponse is exactly such a helper and 109 sites used it. "Limits of the instruments" is rewritten from what was measured	last se tion
F6: tool/main.go errSite misses bare `err` arguments	Fixed in instrument 1 (kind `BARE`). The control needle below fails on the old logic (bare err not reported) and passes on the new	"Contr needle

Round-2 review findings (NO-GO on rework 2), closure in this rework

Finding	Closure	Where
N1: "class complete / no unclosed LEAK" false: class excluded state persisted and later served; sink scan	Class redefined as error-derived text that can reach a client by ANY path. Instrument 1 gained kind `H`:	Class definition, Instru-

Finding	Closure	Where
missed an err inside a helper in a sink (fixture f) and a non-err-named variable in %v (fixture g)	(any call with an err-like identifier argument inside a sink argument) and err-like detection by type/nil-comparison; reviewer fixtures f and g are now found. New instrument 3 (flow scan) lists error-derived text stored in struct fields, map/kv entries, channels and call arguments. New section "Class 4" traces each stored field to its writer, route, gate and scrub. The "no LEAK row remains unclosed" sentence is withdrawn	ments, Class 4, Control needles
N1 specific rows (GetStorageRootStatus, ListStoragePath, SyncSession.ErrorMessage, ScanStatus.Reason + event bus, scrubSecrets and sync_service.go:1057 census rows)	Rows S1 to S5 in Class 4; the census sentence "none reach a response" is corrected	Class 4, Census
N2: unrouted net/http variants counted as live	B7 to B10, B12 to B14, B80 to B83, B213, B214 reclassified LATENT with the routing evidence; totals recounted	Per-site table, Review-base totals
N3: validationText/paramRequiredText rows wrongly "CLOSED"; claim "0 SendErrorResponse sites carry error text"	HEAD status of VALIDATION rows reworded to "NOT CLOSED"; collection_handler.go:335,351, playlist_handler.go:95,188, service_handlers.go:258,298 and input_validation.go:403 added to R4; the claim is replaced	R4, Class 1
N4: configuration_service.go:430,443, loc-		Class 4

Finding	Closure	Where
alization_service.go:849-904, recovery/retry.go:401, webdav_client.go:340-502	Traced in Class 4 rows S6 to S10	
N5: B2 reason, B3 "(unresolved in review base)", ErrRecommendationsNotImplemented	B2 now VALIDATION with the producer cited; B3 reason cites auth_service.go:132; the sentinel text is noted in Class 1	Per-site table, Class 1

Instruments

Both are Go AST programs (go/parser, go/ast), built in a scratch directory; the sources are committed next to this doc as `docs/qa/err_error_class_scan/astscan_sinks.go.txt` and `all_error_calls.go.txt` (`.txt` so no Go tool picks them up). They were run on a clean clone at both bases, with each tree checked out without recursive submodule init.

1. **Rework 3 changes to the sink scan.** Err-like identifiers are now found by type (`error` in parameters, results and `var` declarations) or by comparison with nil, so a variable named `failure` or `cause` counts. Kind `H:` reports any non-`fmt` call that takes an err-like identifier inside a sink argument. Kind `F:` matches any err-like identifier argument of `fmt.Sprint*`.
2. **Sink scan.** For every non-test function it finds calls whose callee name looks like a response sink (names containing json, abortwith, string, write, send, respond, errorresponse, badrequest, servererror, encode, `http.Error`, html, data, emit, broadcast, notify, plus `c.Error`/`AbortWithError`); logger, `fmt.*`, `errors.*`, `strings.*` callees are excluded. An argument is reported when its subtree holds (E) a zero-argument `.Error()` call, (F) `fmt.Sprint*`/`Errorf` with an err-named identifier argument, (T) an identifier that the same function assigned from such an expression (taint, flow-insensitive), or (BARE) an err-named identifier passed as an argument.
3. **Flow scan** (`docs/qa/err_error_class_scan/flows_assign.go.txt`, skips `challenges/`). Lists error-derived text (`err.Error()`, `%v` of an err-like identifier, a variable assigned from either) assigned to a struct field, an index or map key, a key-value pair, a channel send or a call argument, with file, line, function and kind. Its rows are leads for Class 4, not verdicts.

4. **Census.** Every zero-argument `.Error()` call in every non-test file, with its syntactic context (argument of a named call, assignment, return, composite literal / key-value, branch, other). It catches flows instrument 1 cannot see (struct literals returned to a handler, helpers with unusual names).

Control needles (§11.4.201(7)(b), §11.4.273)

- Known positive on the review base: `handlers/error_reporting_handler.go:85` (`http.Error(w, err.Error(), 500)`): found, 1 row, instrument 1. On HEAD that file has 0 rows (closed by CAT-PXF23-R5B).
- Synthetic fixture with four functions (bare err to `SendErrorResponse`; `fmt.Sprintf` into a variable then `c.JSON`; direct `err.Error()` in a map; `err.Error()` only in a log call and an == test): instrument 1 reported exactly the first three (kinds BARE, TEXT-via-taint, TEXT) and nothing for the fourth; instrument 2 reported the direct call (context `kv`), the log argument (`arg-of:log.Printf`) and the comparison (`branch`).
- Reviewer fixtures (round 2): `fixF` (`c.JSON(500, gin.H{"e": scrub(err)})`, helper nested in a sink) and `fixG` (`fmt.Sprintf("failed: %v", failure)` where `failure` is a non-err-named `error`) are both reported by the rebuilt instrument 1 (kinds `H:` and `F:`); good1 to good3 are still reported and bad1 is still silent. The previous build reported neither fixF nor fixG.
- Re-run of instrument 1 on a fresh checkout of each base reproduces the stored outputs: raw 432 rows (review base) and 371 rows (HEAD); 310 and 250 after removing the `strings.`, `sort.`, `bytes.` callee noise. New HEAD rows of kind `H:` are the helpers listed in Class 3 and R4 plus nil-comparison noise (`len(playlists)`, `[]string(p)` and similar, read and discarded as false positives).
- Fabricated token `ZZNEEDLE_NOPE`: 0 hits in all output files (instruments 1, 2 and 3, both bases).
- Cross-check of instrument 2 against `grep -rE '\.Error\(\)'` (non-test .go, catalog-api): review base AST 333 calls vs 329 grep lines; HEAD AST 255 vs 254. Line-set comparison (LC_ALL=C): AST-only lines 0 at both bases; grep-only lines are 1 at the review base and 4 at HEAD (universal_scanner.go:188 in both), and every one was read: they are comments that quote `err.Error()` (universal_scanner.go:188, auth_handler.go:36, client_errors.go:67, utils/response.go:31). The AST count exceeds the grep line count because several lines hold more than one call (5 such lines at each base).
- Taint is flow-insensitive: on HEAD `handlers/user_handler.go:122` is reported as `T:msg` because `msg` is assigned from `err.Error()` later in

the function (line 168); at line 122 `msg` comes from `authClientMessage` (authored text). Read and confirmed a false positive; it is not counted below.

Class definition

A site is in the class when text derived from an `error` value can reach a client (HTTP, WebSocket or SSE response body or header) by ANY path. Routes (rework 3 widens this; the earlier definition excluded server-side state that is persisted and later served): (1) `err.Error()` or `%v/%w/%s` of an error inside a response call; (2) a bare error passed to a helper that serialises it; (3) error text stored in a struct or map that a handler later serialises; (4) error text kept in a variable and used in a response; (5) error text returned from a helper into a response; (6) a helper called inside a sink argument (`gin.H{"e": scrub(err)}`); (7) `%v`/`%s` of an error-typed identifier without `.Error()`, whatever the identifier is called; (8) error text stored in a struct field, map, database row, session or status object and served later by another route (Class 4); (9) error text published on the event bus and forwarded to WebSocket or SSE clients. Sites that only log or branch on text are outside the class; sites that record text in server-side state are IN the class when a route serves it, and are traced in Class 4.

Classes used in the tables: **LEAK** (internal detail can reach the client: SQL/driver text, wrapped `%w` chains, paths, upstream or JWT parse text; "can include", not a demonstrated leak for every row: service and repository strings were read only where stated); **VALIDATION** (authored text that tells the caller which input to fix, or a sentinel's own text; may echo the caller's own input); **LATENT** (leaking code that no non-test caller can reach); **SAFE** (the err value does not reach the client: log-only, fixed-text helper, or a non-client sink); **CLOSED-BY-HELPER** and **CLOSED** are HEAD statuses, not classes.

Classification rules (how the 20 old UNKNOWN rows were resolved)

Rule 1: a 4xx/5xx text produced by a service chain that contains `fmt.Errorf("...: %w", repoErr)` is LEAK (configuration GetWizardProgress, error_reporting GetErrorReport/GetCrashReport, auth Login/Refresh/ChangePassword via auth_service.go wraps, localization Import/Edit). Rule 2: text from a package-level sentinel or a `smartrules.ValidationError` is VALIDATION (playlist, service_handlers, conversion). Rule 3: `ErrLocalRootNotAllowed` text from

`ResolveLocalRoot/CheckLocalRoot` is LEAK (it embeds `cannot resolve path: <OS error>`, local_policy.go:54). Rule 4: a binder (ShouldBindJSON) error on 400 is VALIDATION (mild: names Go struct fields). Rule 5: any other text on a 5xx or a variable status is LEAK.

Class 1: SendErrorResponse family (the review's unlisted class)

`utils.SendErrorResponse(c, status, message, err)` at the review base (`utils/response.go:25-39`): if `err != nil`, `response.Details = err.Error()`, for every status. Bare-err call sites by file (review base):

File	Sites
catalog-api/handlers/media_entity_handler.go	35
catalog-api/handlers/collection_handler.go	15
catalog-api/handlers/comic_pages_handler.go	14
catalog-api/handlers/media_handler.go	10
catalog-api/handlers/stats.go	10
catalog-api/handlers/recommendation_handler.go	6
catalog-api/handlers/search.go	6
catalog-api/handlers/browse.go	5
catalog-api/handlers/challenge.go	5
catalog-api/handlers/media_browse_handler.go	2
catalog-api/middleware/auth.go	1
Total	**109** in 11 files

The review base has 167 SendErrorResponse call sites in non-test code: 109 pass a bare err variable (the table above; `nerr`/`rerr` at comic_pages_handler.go:582,592 are included), 1 passes `err.Error()` as the message (collection_handler.go:335), and 57 pass `nil` or a sentinel (`ErrRecommendationsNotImplemented`, recommendation_handler.go:115,152). The 57 carry no error text and are not rows.

HEAD: `SendErrorResponse` logs err (status, request id) and never sets `Details`; `ErrorResponse.Details` is no longer assigned in `utils/` (grep `Details =`: base 1 hit at response.go:31, HEAD 0), with `TestSendErrorResponse_DoesNotLeakInternalErrorText` and `..._HeadersDoNotLeakErrorText` in `utils/response_test.go`. All 109 rows are therefore CLOSED-BY-HELPER. Residual risk: the helper's `message` argument. At HEAD the `message` argument carries error-derived text at

2 SendErrorResponse sites: collection_handler.go:335 and :351, both `validationText(err)` (authored `ValidationError.Msg`, VALIDATION, listed in R4). The earlier sentence "0 TEXT rows" described only kinds `E:`/`F:`/`T:` and missed the helper kind `H:`; it is withdrawn. The 57 sites that pass `nil` or a sentinel include `ErrRecommendationsNotImplemented` (recommendation_handler.go:115,152): at the review base its own text "recommendation source not implemented" went into `details`; it is authored text, not internal detail (not counted as a row).

Class 2: error text in struct/map literals and indirect flows

Found only by instrument 2 or by taint, not by the sink scan:

Row	File:line (review base)	Flow	Class	HEAD
H1	catalog-api/internal/metrics/health.go:157	`Message: "Database ping failed: " + err.Error()` in a `ComponentHealth` returned by `checkDatabase`, serialised by the unauthenticated `/health/deep` route (registered at `main.go:1107` at base; the comment at `main.go:1052` says "/health and /health/deep stay public")	LEAK	CLOSED (fixed text "Database ping failed", error logged; CAT-PXF23)
H2	catalog-api/internal/monitoring/monitoring.go:251	`"error": err.Error()` in the map built by `HealthChecker.Check`, served by `GinHealthHandler`	LEAK, LATENT (package unwired)	STILL PRESENT, unchanged (see R1)

H1 note: this is the only base instance of the "struct literal" route in `internal/metrics`. Instrument 2 lists exactly two `.Error()` calls in `internal/metrics` and `internal/monitoring` together at the review base

(H1, H2), so the route is exhausted for those packages. user_handler.go:171,176 (`msg := err.Error()` echoed in `details`) are the "variable" route and appear in the per-site table as taint rows.

Class 3: helpers that take an error (class-complete list)

Every non-test function in handlers, middleware, internal/handlers, internal/middleware, utils, internal/metrics and internal/monitoring with an error parameter was read (grep of `func ...(err error`). Result at HEAD:

Helper	Does err text reach the client	Verdict
utils.SendErrorResponse	no (log only)	SAFE
handlers.respondInternal / respondBadBody / stdError (client_errors.go)	no (fixed message / status text)	SAFE
handlers.syncFail (client_errors.go:85)	fixed class text in `details` (syncDetails)	SAFE
handlers.validationText / sentinelText / paramRequiredText	authored text only (ValidationError.Msg, sentinel's own text, " parameter required")	VALIDATION
handlers.authClientMessage (auth_handler.go:39)	sentinel text only	VALIDATION
handlers.respondPDFError, writeLogServiceError, writeFixedLogError	fixed generic texts (already so at the review base)	SAFE
handlers.writeJobError (conversion_handler.go:337)	only `ErrConversionJobState` text	VALIDATION
handlers.rootURLMessage / localRootMessage / rootSpecMessage (scan_handler.go:553-578)	authored policy text; the caller's own path is echoed; OS error after "cannot resolve path" is dropped; URL refusal text is generic	VALIDATION (intentional)
handlers.AliasHandler.fail, middleware.useFallback, internal/handlers.smbErrorStatus	no (log only / fixed text)	SAFE

Helper	Does err text reach the client	Verdict
middleware.clientValidationDetails (input_validation.go:184)	only `validationError.public`	VALIDATION
smartrules.jsonErrText	authored ValidationError message built from the caller's own JSON decode error	VALIDATION
services notifyUser (conversion_service.go:958, sync_service.go:1084)	debug log only; `conversionNotifyHook` has no non-test setter	SAFE (not a client sink)

At the review base the helpers that differed: `writeJobError` sent `err.Error()` for `ErrConversionJobState` (same authored text, VALIDATION), `authClientMessage` and the `client_errors.go` helpers did not exist (grep at the review base: 0 hits for `func authClientMessage` and no `client_errors.go`).

Class 4: error text stored in state and served by another route

Found by instrument 3 (flow scan, below) and then traced by reading the writer, the model's JSON tag, the route registration and the gate. All routes are under the `/api/v1` group of `main.go` (authenticated unless a row says otherwise). Line numbers are HEAD (713a4f380) unless marked.

Row	Field and writer	Served by (route, gate)
S1	`ScanStatus.Reason`: `fail`/`cancel` in universal_scanner.go:505-522 store `scrubSecrets(err.Error(), secrets)`	`GET /scans`, `GET /scans/:job_id` (`scan_handler.go:539` `"reason":` `RegisterScanRoutes`, storage_rou authenticated user. Also `publish` versal_scanner.go:161-194, payl -> `EventBusBridge` (internal/hand bus_bridge.go:74,96,189-190; wi main.go:681,725) -> a `notificati` broadcast to all WebSocket clie

Row	Field and writer	Served by (route, gate)
		(`BroadcastToClients`; the websoc thenticated, main.go:1151)
S2	`GetStorageRootStatus` (scan_handler.go:332,339,345): `resp["error"] = services.ScrubErrorFor(err, root)`, HTTP 200	`/storage-roots/:id/status`, no R (storage_routes.go:18): any auth
S3	`ListStoragePath` (scan_handler.go:378,384,401): same call	`/storage-roots/:id/list`, admin age_routes.go:14)
S4	`SyncSession.ErrorMessage`: `handleSyncError` (sync_service.go:1053-1066) stores raw `syncError.Error()`; chains such as "failed to get WebDAV client / scan local files / list remote files: %w" (models/user.go:1229, json `error_message,omitempty`)	`GET /sync/sessions`, `GET /sync/s` `GET /sync/history` (main.go:165 caller's own sessions, no admin
S5	`ConversionJob.ErrorMessage` (models/file.go:158): conversion_service.go:884-885 stores `sanitizeConversionError(err.Error())` (reduces paths to base names)	`GET /conversion/jobs`, `GET /conv` (main.go:1410-1411): any auther

S6	`WizardStepValidation.Errors[...]` (configuration_service.go:420,430,443): the three validators that can return an error return fixed authored texts ("path must be absolute", "invalid email format"), the database and network validators return nil; "%s is required" uses the field label	`POST /configuration/wizard/ste`
		`validate` ->`ConfigurationServiceAdapter.Va`
		(handlers/service_adapters.go:7
		(main.go:1495), gated by `system`

S7	`ConfigurationImportResult.ValidationErrors/SkippedSettings` (internal/services/localization_service.go:849,857,877,886,895,904)	no route: `LocalizationHandlers` only in tests (`NewLocalizationHa`

Row	Field and writer	Served by (route, gate)
		test callers; main.go registers no `configuration` path)
S8	`BatchResult.Errors`, `SyncResult.Errors` (services/webdav_client.go:340,361,429,450,482,502)	no non-test caller of `UploadBatch` `SyncDirectory` outside the file (g dav_client_test.go)
S9	`retry` result `Error` (internal/recovery/retry.go:401)	`internal/recovery` has no non-te `internal/recovery"`: 0 hits; "digi al.vasic.recovery" in internal/mo is a name string)
S10	`liveevidence.Record.ResolveError` (internal/liveevidence/liveevidence.go:158)	written to evidence files by Go t importer
S11	`log_management_service.logError` (log_management_service.go:170,194,204): `message` is formatted from `%v` of errors	the function only calls the struct (lines 724-732)
S12	`notifyUser` (conversion_service.go, sync_service.go), `logSyncError`/`logSkippedFile`	log or debug output only

Instrument 3 reports 362 flow rows at HEAD (402 at the review base; `challenges/` is skipped, see R3). The field, key-value and call-argument rows of the files named in S1 to S12 were each traced above. The remaining rows (mostly constant or authored text assigned to fields, and calls into logging helpers) were NOT each traced one by one: UNCONFIRMED whether any of them reaches a client through a route this document did not look at (see Limits).

HEAD residual (everything still in the class at 713a4f380)

Id	Site	Class	No
R1	catalog-api/internal/monitoring/monitoring.go:251 `"error": err.Error()` (GinHealthHandler)	LATENT-OPEN	th ha tes th co mo wi sa let

Id	Site	Class	No
R2	OCU-CUDA-Sidecar/internal/server/server.go:110,115,128,133,150,154,159,172,176,180,185,198,203,216,221 (15 sites, `http.Error(w, err.Error()...)` and `"bad request: "+err.Error()`)	LEAK	a s mo 15 ba th sc do in- ca `Li` an `se` wh pr re un en CO
R3	catalog-api/handlers/challenge.go and the `challenges/` package: 155 `.Error()` calls (74 `CreateResult`, 20 `image_quality_final.go`, 16 `image_quality.go`, 13 `ch131_middleware_challenges.go`, ...) feed challenge result records that `ChallengeHandler.GetResults`, `RunChallenge` and `RunAll` return as JSON (`handlers/challenge.go:115,185`); `POST` run routes are admin-gated (`main.go:1670-1672`), the gating of `GET /api/v1/challenges/results` was not established	UNCONFIRMED	no pe cic res op
R4	VALIDATION sentinel/authored sites: playlist_handler.go:95,99,188,273,320; collection_handler.go:335,351 (`validationText(err)` as the SendErrorResponse message); service_handlers.go:199,258,298 (`paramRequiredText(err)`); conversion_handler.go:344; scan_handler.go:178,186,194,448; middleware/input_validation.go:403 (`clientValidationDetails(err)`)	VALIDATION	au sc ec ca pa ro clo sti th
R5			re fix

Id	Site	Class	No
	`handlers/scan_handler.go` localRootMessage keeps `msg[:i] + ": cannot resolve path"`	VALIDATION	the ser the ca
R6	Class 4 rows S1, S2, S4, S5: `ScanStatus.Reason` (also on the WebSocket broadcast), `GetStorageRootStatus` `error`, `SyncSession.ErrorMessage`, `ConversionJob.ErrorMessage`	LEAK (partly scrubbed)	op ser the us ow fix for ro ful for the ch thi

Everything else the sink scan reports at HEAD is SAFE by the helper table (111 `SendErrorResponse` + 16 `respondInternal/respondBadBody` + 5 `respondPDFError` + 5 `writeLogServiceError` + 5 `writeJobError` + 2 `c.Error` bare-err rows; the `c.Error` rows are ErrorTypePrivate, which ErrorHandler answers with a fixed message at HEAD: `internal/middleware/middleware.go:97-100`).

Census (instrument 2, all non-test code under catalog-api)

Group	Review base	HEAD
production code	175	97
challenges/ (harness, see R3)	155	155
internal/tests (test helpers)	3	3
total	333	255

HEAD production-code contexts (97): branching on text (`strings.Contains` 42, `strings.ToLower` 7, `strings.TrimPrefix` 2, `branch` 6) = 57, read at the branch: the compared text does not itself go to a response (stored-then-served flows are in Class 4); assignment 14, return 9, key-value 6, scrub/log helpers (`scrubSecrets` 5, `logSkippedFile` 2, `sanitizeConversionError` 1, `panic` 1, `paramRequiredRE.FindString` 1),

`fmt.Sprintf` 1 (sync_service.go:1066, notifyUser, SAFE). `scrubSecrets` at universal_scanner.go:510,520 stores the scrubbed text in `ScanStatus.Reason`, which IS served (row S1); sync_service.go:1057-1058 `handleSyncError` stores the raw text in `SyncSession.ErrorMessage`, which IS served (row S4). The earlier sentences that none of these reach a response are withdrawn. The 14 assign + 9 return + 6 kv rows were read with the helper table above; the two that carry text to a body are the sentinel rows in R4 and the unwired R1.

Review-base totals (instrument 1, 226 rows after removing `strings.*` callee noise) plus H1/H2

Class	SendErrorResponse family	other sinks
LEAK	109	49
VALIDATION	1	33
LATENT	0	15
SAFE	0	19
total	110	116

HEAD status of the 226 rows: CLOSED-BY-HELPER 109; CLOSED (err text no longer sent) 64 (49 LEAK + 15 LATENT); VALIDATION rows where `err.Error()` is no longer sent but authored text still is (NOT CLOSED) 32; unchanged (SAFE 19, VALIDATION 2) 21. This count covers the direct-sink rows only. It is NOT a claim that no leak remains: Class 4 rows S1 to S5 are still served at HEAD with credential-only or path-only scrubbing, and R1 to R3 are open.

Per-site table (review base 9ed6577b, with HEAD status)

Row ids B1.. are sorted by file and line. `st` is the status argument when the call has one.

#	File:line	Function	Sink	st
B1	catalog-api/handlers/asset_handler.go:91	RequestAsset	c.JSON	http.S Reque

#	File:line	Function	Sink	st
B2	catalog-api/handlers/auth_handler.go:36	LoginGin	c.JSON	http.S bidde
B3	catalog-api/handlers/auth_handler.go:39	LoginGin	c.JSON	http.S thoriz
B4	catalog-api/handlers/auth_handler.go:59	RefreshToken-Gin	c.JSON	http.S thoriz
B5	catalog-api/handlers/auth_handler.go:76	LogoutGin	c.JSON	http.S ternal ror
B6	catalog-api/handlers/auth_handler.go:203	RegisterGin	c.JSON	http.S Reque
B7	catalog-api/handlers/auth_handler.go:321	Login	http.Error	http.S thoriz

#	File:line	Function	Sink	st
B8	catalog-api/handlers/auth_handler.go:346	RefreshToken	http.Error	http.S thoriz
B9	catalog-api/handlers/auth_handler.go:368	Logout	http.Error	http.S ternal ror
B10	catalog-api/handlers/auth_handler.go:390	LogoutAll	http.Error	http.S ternal ror
B11	catalog-api/handlers/auth_handler.go:440	ChangePass-word	http.Error	http.S Reque
B12	catalog-api/handlers/auth_handler.go:446	ChangePass-word	http.Error	http.S Reque

#	File:line	Function	Sink	st
B13	catalog-api/handlers/auth_handler.go:468	GetActiveSessions	http.Error	http.Sternalror
B14	catalog-api/handlers/auth_handler.go:507	DeactivateSession	http.Error	http.Sternalror
B15	catalog-api/handlers/browse.go:41	GetStorageRoots	utils.SendErrorResponse	http.Sternalror
B16	catalog-api/handlers/browse.go:120	BrowseDirectory	utils.SendErrorResponse	http.Sternalror
B17	catalog-api/handlers/browse.go:164	GetFileInfo	utils.SendErrorResponse	http.Sternalror
B18	catalog-api/handlers/browse.go:217	GetDirectorySizes	utils.SendErrorResponse	http.Sternalror
B19	catalog-api/handlers/browse.go:270	GetDirectoryDuplicates	utils.SendErrorResponse	http.Sternalror
B20		RunChallenge	utils.SendErrorResponse	

#	File:line	Function	Sink	st
	catalog-api/handlers/challenge.go:103			http.S Found
B21	catalog-api/handlers/challenge.go:106	RunChallenge	utils.SendErrorResponse	http.S ternal ror
B22	catalog-api/handlers/challenge.go:166	RunAll	utils.SendErrorResponse	http.S ternal ror
B23	catalog-api/handlers/challenge.go:234	RunByCategory	utils.SendErrorResponse	http.S Found
B24	catalog-api/handlers/challenge.go:237	RunByCategory	utils.SendErrorResponse	http.S ternal ror
B25	catalog-api/handlers/collection_handler.go:100	loadAccessible	utils.SendErrorResponse	http.S Reque
B26	catalog-api/handlers/collection_handler.go:105	loadAccessible	utils.SendErrorResponse	http.S Found
B27	catalog-api/handlers/collection_handler.go:160	ListCollections	utils.SendErrorResponse	http.S ternal ror

#	File:line	Function	Sink	st
B28	catalog-api/handlers/collection_handler.go:269	GetCollection	utils.SendErrorResponse	http.Sternalror
B29	catalog-api/handlers/collection_handler.go:315	ListCollectionItems	utils.SendErrorResponse	http.Sternalror
B30	catalog-api/handlers/collection_handler.go:335	parseSmartRules	utils.SendErrorResponse	http.SReque
B31	catalog-api/handlers/collection_handler.go:337	parseSmartRules	utils.SendErrorResponse	http.Sternalror
B32	catalog-api/handlers/collection_handler.go:353	verifyRules	utils.SendErrorResponse	-
B33	catalog-api/handlers/collection_handler.go:374	CreateCollection	utils.SendErrorResponse	http.SReque
B34	catalog-api/handlers/collection_handler.go:429	CreateCollection	utils.SendErrorResponse	http.Sternalror

#	File:line	Function	Sink	st
B35	catalog-api/handlers/collection_handler.go:452	CreateCollection	utils.SendErrorResponse	http.Sternalror
B36	catalog-api/handlers/collection_handler.go:476	UpdateCollection	utils.SendErrorResponse	http.SReque
B37	catalog-api/handlers/collection_handler.go:480	UpdateCollection	utils.SendErrorResponse	http.SReque
B38	catalog-api/handlers/collection_handler.go:539	UpdateCollection	utils.SendErrorResponse	http.Sternalror
B39	catalog-api/handlers/collection_handler.go:561	DeleteCollection	utils.SendErrorResponse	http.SFound
B40	catalog-api/handlers/collection_handler.go:564	DeleteCollection	utils.SendErrorResponse	http.Sternalror
B41	catalog-api/handlers/comic_pages_handler.go:106	resolveArchive	utils.SendErrorResponse	http.SReque
B42	catalog-api/handlers/comic_pages_handler.go:113	resolveArchive	utils.SendErrorResponse	http.SFound

#	File:line	Function	Sink	st
B43	catalog-api/handlers/comic_pages_handler.go:116	resolveArchive	utils.SendErrorResponse	http.S ternal ror
B44	catalog-api/handlers/comic_pages_handler.go:434	ListComicPages	utils.SendErrorResponse	http.S ternal ror
B45	catalog-api/handlers/comic_pages_handler.go:467	GetComicPage	utils.SendErrorResponse	http.S Reque
B46	catalog-api/handlers/comic_pages_handler.go:478	GetComicPage	utils.SendErrorResponse	http.S ternal ror
B47	catalog-api/handlers/comic_pages_handler.go:492	GetComicPage	utils.SendErrorResponse	http.S ternal ror
B48	catalog-api/handlers/comic_pages_handler.go:515	listCBRPages	utils.SendErrorResponse	http.S ternal ror
B49	catalog-api/handlers/comic_pages_handler.go:522	listCBRPages	utils.SendErrorResponse	http.S ternal ror
B50	catalog-api/handlers/comic_pages_handler.go:552	getCBRPage	utils.SendErrorResponse	http.S ternal ror

#	File:line	Function	Sink	st
B51	catalog-api/handlers/comic_pages_handler.go:559	getCBRPage	utils.SendErrorResponse	http.Sternalror
B52	catalog-api/handlers/comic_pages_handler.go:570	getCBRPage	utils.SendErrorResponse	http.Sternalror
B53	catalog-api/handlers/comic_pages_handler.go:582	getCBRPage	utils.SendErrorResponse	http.Sternalror
B54	catalog-api/handlers/comic_pages_handler.go:592	getCBRPage	utils.SendErrorResponse	http.Sternalror
B55	catalog-api/handlers/configuration_handler.go:57	GetWizardStep	http.Error	http.SFound
B56	catalog-api/handlers/configuration_handler.go:77	ValidateWizardStep	http.Error	http.Sternalror
B57	catalog-api/handlers/configuration_handler.go:102	SaveWizardProgress	http.Error	http.Sternalror
B58	catalog-api/handlers/configuration_handler.go:121	GetWizardProgress	http.Error	http.SFound
B59			http.Error	

#	File:line	Function	Sink	st
	catalog-api/handlers/configuration_handler.go:145	CompleteWizard		http.Sternalror
B60	catalog-api/handlers/configuration_handler.go:173	GetConfiguration	http.Error	http.Sternalror
B61	catalog-api/handlers/configuration_handler.go:203	TestConfiguration	http.Error	http.Sternalror
B62	catalog-api/handlers/conversion_handler.go:73	CreateJob	c.JSON	http.SReque
B63	catalog-api/handlers/conversion_handler.go:99	GetJob	writeJobError	-
B64	catalog-api/handlers/conversion_handler.go:185	CancelJob	writeJobError	-
B65	catalog-api/handlers/conversion_handler.go:246	DeleteJob	writeJobError	-
B66	catalog-api/handlers/conversion_handler.go:282	RetryJob	writeJobError	-
B67	catalog-api/handlers/conversion_handler.go:307	DownloadJobFile	writeJobError	-
B68	catalog-api/handlers/conversion_handler.go:344	writeJobError	c.JSON	http.Sflict

#	File:line	Function	Sink	st
B69	catalog-api/handlers/error_reporting_handler.go:85	ReportError	http.Error	http.Sternalror
B70	catalog-api/handlers/error_reporting_handler.go:123	ReportCrash	http.Error	http.Sternalror
B71	catalog-api/handlers/error_reporting_handler.go:152	GetErrorReport	http.Error	http.SFound
B72	catalog-api/handlers/error_reporting_handler.go:181	GetCrashReport	http.Error	http.SFound
B73	catalog-api/handlers/error_reporting_handler.go:205	ListErrorReports	http.Error	http.Sternalror
B74	catalog-api/handlers/error_reporting_handler.go:232	ListCrashReports	http.Error	http.Sternalror
B75	catalog-api/handlers/error_reporting_handler.go:273	UpdateErrorStatus	http.Error	http.Sternalror
B76	catalog-api/handlers/error_reporting_handler.go:310	UpdateCrashStatus	http.Error	http.Sternalror
B77	catalog-api/handlers/error_reporting_handler.go:332	GetErrorStatistics	http.Error	http.Sternalror
B78	catalog-api/handlers/error_reporting_handler.go:355	GetCrashStatistics	http.Error	http.Sternalror
B79	catalog-api/handlers/error_reporting_handler.go:378	GetSystemHealth	http.Error	http.Sternalror
B80	catalog-api/handlers/error_reporting_handler.go:407	UpdateConfiguration	http.Error	http.Sternalror

#	File:line	Function	Sink	st
B81	catalog-api/handlers/error_reporting_handler.go:432	GetConfiguration	http.Error	http.Sternalror
B82	catalog-api/handlers/error_reporting_handler.go:469	CleanupOldReports	http.Error	http.Sternalror
B83	catalog-api/handlers/error_reporting_handler.go:495	ExportReports	http.Error	http.Sternalror
B84	catalog-api/handlers/log_management_handler.go:222	GetLogEntries	writeLogServiceError	-
B85	catalog-api/handlers/log_management_handler.go:254	CreateLogShare	writeLogServiceError	-
B86	catalog-api/handlers/log_management_handler.go:322	RevokeLogShare	writeLogServiceError	-

#	File:line	Function	Sink	st
B87	catalog-api/handlers/log_management_handler.go:355	ExportLogs	writeLogServiceError	-
B88	catalog-api/handlers/log_management_handler.go:475	AnalyzeLogs	writeLogServiceError	-
B89	catalog-api/handlers/media_browse_handler.go:83	SearchMedia	utils.SendErrorResponse	http.S ternal ror
B90	catalog-api/handlers/media_browse_handler.go:106	GetMediaStats	utils.SendErrorResponse	http.S ternal ror
B91	catalog-api/handlers/media_browse_handler.go:127	GetMediaStats	c.Error	-
B92	catalog-api/handlers/media_browse_handler.go:158	GetMediaStats	c.Error	-
B93	catalog-api/handlers/media_entity_handler.go:77	SearchEntities	utils.SendErrorResponse	http.S Reque
B94	catalog-api/handlers/media_entity_handler.go:85	SearchEntities	utils.SendErrorResponse	http.S ternal ror

#	File:line	Function	Sink	st
B95	catalog-api/handlers/media_entity_handler.go:227	ListEntities	utils.SendErrorResponse	http.S Reque
B96	catalog-api/handlers/media_entity_handler.go:239	ListEntities	utils.SendErrorResponse	http.S ternal ror
B97	catalog-api/handlers/media_entity_handler.go:260	GetEntity	utils.SendErrorResponse	http.S Reque
B98	catalog-api/handlers/media_entity_handler.go:266	GetEntity	utils.SendErrorResponse	http.S Found
B99	catalog-api/handlers/media_entity_handler.go:306	GetEntityChildren	utils.SendErrorResponse	http.S Reque
B100	catalog-api/handlers/media_entity_handler.go:315	GetEntityChildren	utils.SendErrorResponse	http.S ternal ror
B101	catalog-api/handlers/media_entity_handler.go:336	GetEntityFiles	utils.SendErrorResponse	http.S Reque
B102	catalog-api/handlers/media_entity_handler.go:342	GetEntityFiles	utils.SendErrorResponse	http.S ternal ror

#	File:line	Function	Sink	st
B103	catalog-api/handlers/media_entity_handler.go:358	GetEntityMetadata	utils.SendErrorResponse	http.S Reque
B104	catalog-api/handlers/media_entity_handler.go:364	GetEntityMetadata	utils.SendErrorResponse	http.S ternal ror
B105	catalog-api/handlers/media_entity_handler.go:379	GetEntityDuplicates	utils.SendErrorResponse	http.S Reque
B106	catalog-api/handlers/media_entity_handler.go:385	GetEntityDuplicates	utils.SendErrorResponse	http.S Found
B107	catalog-api/handlers/media_entity_handler.go:391	GetEntityDuplicates	utils.SendErrorResponse	http.S ternal ror
B108	catalog-api/handlers/media_entity_handler.go:415	GetEntityTypes	utils.SendErrorResponse	http.S ternal ror
B109	catalog-api/handlers/media_entity_handler.go:457	BrowseByType	utils.SendErrorResponse	http.S Reque
B110	catalog-api/handlers/media_entity_handler.go:463	BrowseByType	utils.SendErrorResponse	http.S ternal ror

#	File:line	Function	Sink	st
B111	catalog-api/handlers/media_entity_handler.go:496	RefreshEntityMetadata	utils.SendErrorResponse	http.S Reque
B112	catalog-api/handlers/media_entity_handler.go:1158	entityMustExist	utils.SendErrorResponse	http.S Found
B113	catalog-api/handlers/media_entity_handler.go:1160	entityMustExist	utils.SendErrorResponse	http.S ternal ror
B114	catalog-api/handlers/media_entity_handler.go:1173	UpdateUserMetadata	utils.SendErrorResponse	http.S Reque
B115	catalog-api/handlers/media_entity_handler.go:1186	UpdateUserMetadata	utils.SendErrorResponse	http.S Reque
B116	catalog-api/handlers/media_entity_handler.go:1217	UpdateUserMetadata	utils.SendErrorResponse	http.S ternal ror
B117	catalog-api/handlers/media_entity_handler.go:1246	StreamEntity	utils.SendErrorResponse	http.S Reque
B118	catalog-api/handlers/media_entity_handler.go:1260	StreamEntity	utils.SendErrorResponse	http.S Found

#	File:line	Function	Sink	st
B119	catalog-api/handlers/media_entity_handler.go:1263	StreamEntity	utils.SendErrorResponse	http.Sternalror
B120	catalog-api/handlers/media_entity_handler.go:1283	DownloadEntity	utils.SendErrorResponse	http.SReque
B121	catalog-api/handlers/media_entity_handler.go:1291	DownloadEntity	utils.SendErrorResponse	http.SFound
B122	catalog-api/handlers/media_entity_handler.go:1320	DownloadEntity	utils.SendErrorResponse	http.SFound
B123	catalog-api/handlers/media_entity_handler.go:1323	DownloadEntity	utils.SendErrorResponse	http.Sternalror
B124	catalog-api/handlers/media_entity_handler.go:1343	GetInstallInfo	utils.SendErrorResponse	http.SReque
B125	catalog-api/handlers/media_entity_handler.go:1349	GetInstallInfo	utils.SendErrorResponse	http.SFound
B126	catalog-api/handlers/media_entity_handler.go:1369	GetInstallInfo	utils.SendErrorResponse	http.SFound

#	File:line	Function	Sink	st
B127	catalog-api/handlers/media_entity_handler.go:1402	ListDuplicateGroups	utils.SendErrorResponse	http.Sternalror
B128	catalog-api/handlers/media_handler.go:46	GetMediaByID	utils.SendErrorResponse	http.SReque
B129	catalog-api/handlers/media_handler.go:87	GetMediaByID	utils.SendErrorResponse	http.SFound
B130	catalog-api/handlers/media_handler.go:89	GetMediaByID	utils.SendErrorResponse	http.Sternalror
B131	catalog-api/handlers/media_handler.go:148	UpdateWatchProgress	utils.SendErrorResponse	http.SReque
B132	catalog-api/handlers/media_handler.go:155	UpdateWatchProgress	utils.SendErrorResponse	http.SReque
B133	catalog-api/handlers/media_handler.go:205	UpdateWatchProgress	utils.SendErrorResponse	http.Sternalror
B134	catalog-api/handlers/media_handler.go:238	UpdateFavoriteStatus	utils.SendErrorResponse	http.SReque

#	File:line	Function	Sink	st
B135	catalog-api/handlers/media_handler.go:245	UpdateFavoriteStatus	utils.SendErrorResponse	http.S Reque
B136	catalog-api/handlers/media_handler.go:267	UpdateFavoriteStatus	utils.SendErrorResponse	http.S ternal ror
B137	catalog-api/handlers/media_handler.go:274	UpdateFavoriteStatus	utils.SendErrorResponse	http.S ternal ror
B138	catalog-api/handlers/pdf_pages_handler.go:101	resolvePDF	respondPDFError	-
B139	catalog-api/handlers/pdf_pages_handler.go:282	ListPdfPages	respondPDFError	-
B140	catalog-api/handlers/pdf_pages_handler.go:289	ListPdfPages	respondPDFError	-
B141	catalog-api/handlers/pdf_pages_handler.go:322	GetPdfPage	respondPDFError	-

#	File:line	Function	Sink	st
B142	catalog-api/handlers/pdf_pages_handler.go:329	GetPdfPage	respondPDFError	-
B143	catalog-api/handlers/playback_handler.go:79	StartSession	c.JSON	http.S Reque
B144	catalog-api/handlers/playback_handler.go:96	StartSession	c.JSON	http.S ternal ror
B145	catalog-api/handlers/playback_handler.go:106	ProgressSession	c.JSON	http.S Reque
B146	catalog-api/handlers/playback_handler.go:118	ProgressSession	c.JSON	http.S ternal ror
B147	catalog-api/handlers/playback_handler.go:128	EndSession	c.JSON	http.S Reque

#	File:line	Function	Sink	st
B148	catalog-api/handlers/playback_handler.go:143	EndSession	c.JSON	http.Sternalror
B149	catalog-api/handlers/playback_handler.go:170	GetProgressForEntity	c.JSON	http.Sternalror
B150	catalog-api/handlers/playback_handler.go:193	ListHistoryForEntity	c.JSON	http.Sternalror
B151	catalog-api/handlers/playlist_handler.go:95	CreatePlaylist	c.JSON	http.SReque
B152	catalog-api/handlers/playlist_handler.go:99	CreatePlaylist	c.JSON	http.Sflict
B153	catalog-api/handlers/playlist_handler.go:188	UpdatePlaylist	c.JSON	http.SReque

#	File:line	Function	Sink	st
B154	catalog-api/handlers/playlist_handler.go:273	AddItem	c.JSON	http.Sflict
B155	catalog-api/handlers/playlist_handler.go:320	RemoveItem	c.JSON	http.Sflict
B156	catalog-api/handlers/recommendation_handler.go:54	GetSimilarItems	utils.SendErrorResponse	http.SReque
B157	catalog-api/handlers/recommendation_handler.go:76	GetSimilarItems	utils.SendErrorResponse	http.Sternalror
B158	catalog-api/handlers/recommendation_handler.go:93	GetSimilarItems	utils.SendErrorResponse	http.Sternalror
B159	catalog-api/handlers/recommendation_handler.go:115	GetTrendingItems	utils.SendErrorResponse	http.StImple
B160			utils.SendErrorResponse	

#	File:line	Function	Sink	st
	catalog-api/handlers/recommendation_handler.go:135	GetPersonalizedRecommendations		http.S Reque
B161	catalog-api/handlers/recommendation_handler.go:152	GetPersonalizedRecommendations	utils.SendErrorResponse	http.S tImple
B162	catalog-api/handlers/scan_handler.go:153	CreateStorageRoot	c.JSON	http.S Reque
B163	catalog-api/handlers/scan_handler.go:177	CreateStorageRoot	c.JSON	http.S Reque
B164	catalog-api/handlers/scan_handler.go:185	CreateStorageRoot	c.JSON	http.S Reque
B165			c.JSON	

#	File:line	Function	Sink	st
	catalog-api/handlers/scan_handler.go:193	CreateStorage-Root		http.S bidde
B166	catalog-api/handlers/scan_handler.go:227	CreateStorage-Root	c.JSON	http.S ternal ror
B167	catalog-api/handlers/scan_handler.go:247	GetStorage-Roots	c.JSON	http.S ternal ror
B168	catalog-api/handlers/scan_handler.go:427	QueueScan	c.JSON	http.S Reque
B169	catalog-api/handlers/scan_handler.go:441	QueueScan	c.JSON	http.S Found
B170	catalog-api/handlers/scan_handler.go:447	QueueScan	c.JSON	http.S bidde
B171	catalog-api/handlers/scan_handler.go:463	QueueScan	c.JSON	http.S viceU
B172	catalog-api/handlers/search.go:100	SearchFiles	utils.SendErrorResponse	http.S Reque
B173	catalog-api/handlers/search.go:108	SearchFiles	utils.SendErrorResponse	http.S Reque
B174		SearchFiles	utils.SendErrorResponse	

#	File:line	Function	Sink	st
	catalog-api/handlers/search.go:155			http.Sternalror
B175	catalog-api/handlers/search.go:258	SearchDuplicates	utils.SendErrorResponse	http.Sternalror
B176	catalog-api/handlers/search.go:287	AdvancedSearch	utils.SendErrorResponse	http.SReque
B177	catalog-api/handlers/search.go:328	AdvancedSearch	utils.SendErrorResponse	http.Sternalror
B178	catalog-api/handlers/service_handlers.go:199	CreateReport	c.JSON	http.SReque
B179	catalog-api/handlers/service_handlers.go:258	GetUsageReport	c.JSON	http.SReque
B180			c.JSON	

#	File:line	Function	Sink	st
	catalog-api/handlers/service_handlers.go:298	GetPerformanceReport		http.S Reque
B181	catalog-api/handlers/stats.go:42	GetOverallStats	utils.SendErrorResponse	http.S ternal ror
B182	catalog-api/handlers/stats.go:82	GetSmbRootStats	utils.SendErrorResponse	http.S Found
B183	catalog-api/handlers/stats.go:85	GetSmbRootStats	utils.SendErrorResponse	http.S ternal ror
B184	catalog-api/handlers/stats.go:119	GetFileTypeStats	utils.SendErrorResponse	http.S ternal ror
B185	catalog-api/handlers/stats.go:146	GetSizeDistribution	utils.SendErrorResponse	http.S ternal ror
B186	catalog-api/handlers/stats.go:173	GetDuplicateStats	utils.SendErrorResponse	http.S ternal ror
B187			utils.SendErrorResponse	

#	File:line	Function	Sink	st
	catalog-api/handlers/stats.go:214	GetTopDuplicateGroups		http.Sternalror
B188	catalog-api/handlers/stats.go:248	GetAccessPatterns	utils.SendErrorResponse	http.Sternalror
B189	catalog-api/handlers/stats.go:282	GetGrowthTrends	utils.SendErrorResponse	http.Sternalror
B190	catalog-api/handlers/stats.go:321	GetScanHistory	utils.SendErrorResponse	http.Sternalror
B191	catalog-api/handlers/sync_handler.go:40	CreateEndpoint	c.JSON	http.SReque
B192	catalog-api/handlers/sync_handler.go:52	CreateEndpoint	c.JSON	-
B193	catalog-api/handlers/sync_handler.go:69	GetUserEndpoints	c.JSON	http.Sternalror
B194	catalog-api/handlers/sync_handler.go:103	GetEndpoint	c.JSON	-
B195			c.JSON	

#	File:line	Function	Sink	st
	catalog-api/handlers/sync_handler.go:126	UpdateEndpoint		http.S Reque
B196	catalog-api/handlers/sync_handler.go:142	UpdateEndpoint	c.JSON	-
B197	catalog-api/handlers/sync_handler.go:171	DeleteEndpoint	c.JSON	-
B198	catalog-api/handlers/sync_handler.go:202	StartSync	c.JSON	-
B199	catalog-api/handlers/sync_handler.go:234	GetUserSessions	c.JSON	http.S ternal ror
B200	catalog-api/handlers/sync_handler.go:268	GetSession	c.JSON	-
B201	catalog-api/handlers/sync_handler.go:288	ScheduleSync	c.JSON	http.S Reque
B202	catalog-api/handlers/sync_handler.go:304	ScheduleSync	c.JSON	-
B203	catalog-api/handlers/sync_handler.go:341	GetSyncStatistics	c.JSON	

#	File:line	Function	Sink	st
				http.S ternal ror
B204	catalog-api/handlers/ sync_handler.go:371	CleanupOldSes- sions	c.JSON	http.S ternal ror
B205	catalog-api/handlers/ user_handler.go:121	CreateUser	http.Error	http.S Reque
B206	catalog-api/handlers/ user_handler.go:171	CreateUser	http.Error	http.S flict
B207	catalog-api/handlers/ user_handler.go:176	CreateUser	http.Error	http.S ternal ror
B208	catalog-api/handlers/ user_handler.go:624	ResetPassword	http.Error	http.S Reque
B209		ResetPassword	http.Error	

#	File:line	Function	Sink	st
	catalog-api/handlers/user_handler.go:630			http.Sternalror
B210	catalog-api/handlers/user_handler.go:694	LockAccount	http.Error	http.Sternalror
B211	catalog-api/handlers/user_handler.go:738	UnlockAccount	http.Error	http.Sternalror
B212	catalog-api/internal/handlers/copy.go:249	CopyToStorage	c.JSON	http.SReque
B213	catalog-api/internal/handlers/localization_handlers.go:301	ImportConfiguration	h.sendError	http.SReque
B214	catalog-api/internal/handlers/localization_handlers.go:365	EditConfiguration	h.sendError	http.SReque
B215	catalog-api/internal/handlers/smb_discovery.go:86	DiscoverShares	c.JSON	http.SReque

#	File:line	Function	Sink	st
B216	catalog-api/internal/handlers/smb_discovery.go:124	TestConnection	c.JSON	http.S Reque
B217	catalog-api/internal/handlers/smb_discovery.go:176	BrowseShare	c.JSON	http.S Reque
B218	catalog-api/internal/middleware/middleware.go:81	ErrorHandler	c.JSON	http.S Reque
B219	catalog-api/internal/middleware/middleware.go:86	ErrorHandler	c.JSON	http.S ternal ror
B220	catalog-api/internal/smartrules/smartrules.go:207	parseList	jsonErrText	-
B221	catalog-api/main.go:1306	main	c.JSON	

#	File:line	Function	Sink	st
				http.S ternal ror
B222	catalog-api/main.go:1337	main	c.JSON	http.S ternal ror
B223	catalog-api/middleware/ auth.go:167	RequireAuth	utils.SendErrorResponse	http.S thoriz
B224	catalog-api/middleware/in- put_validation.go:360	InputValidation	c.JSON	http.S Reque
B225	catalog-api/services/con- version_service.go:903	handleConver- sionError	s.notifyUser	-
B226	catalog-api/services/ sync_service.go:1066	handleSyncEr- ror	s.notifyUser	-

| H1 | catalog-api/internal/metrics/health.go:157 | checkDatabase | struct literal | - | `"Database ping failed: " + err.Error()` | LEAK | unauthenticated /health/deep body | CLOSED (fixed text, error logged) |
| | H2 | catalog-api/internal/monitoring/monitoring.go:251 | Check | map literal | - | `err.Error()` | LEAK (latent) | package unwired | STILL PRESENT (R1) |

Limits of the instruments (measured, UNCONFIRMED where stated)

- Type information is heuristic: any zero-argument `.Error()` matches, including non-error types (`writer.Error()` of encoding/csv appears in the census; read, not a response); err-like identifiers are found by declared type `error` or a nil comparison, not by the type checker, so an `error` value obtained from a multi-value call and never compared with nil, a field of type `error`, or an interface method result is missed (round-2 blind spot, partly closed).
- Sink detection is by callee name; a response writer with a name outside the list is missed by instrument 1 and is caught only if instrument 2 sees an `.Error()` inside it. Error text built without `.Error()`, `%v` of an err-named identifier, or an err-named identifier argument (for instance a variable named `cause` or `e` passed to `fmt.Sprintf`) is not found by either instrument: UNCONFIRMED whether any exists.
- Blind spots found by the round-2 review and closed only as far as stated: helper-in-sink (`H:`), non-err-named variable in `%v` (`F:`). Still NOT covered: error text round-tripped through the database or a file and served by a different route than the writer's, event-bus consumers other than `EventBusBridge`, text built by `errors.Unwrap`/`errors.Join` chains and then formatted with `%s` of a non-error value, `fmt.Errorf` results returned to a handler that sends them (covered only when the handler's sink argument shows an err-like identifier), and routes registered through dynamic mux patterns. Class 4 is a trace of the fields instrument 3 surfaced, not a proof that no other field is served.
- Taint is flow-insensitive within a function (one false positive found and read, user_handler.go:122 at HEAD). Taint through struct fields, other functions or channels is not followed; the census catches the `.Error()` call itself but not where the stored text is later sent.
- BARE reports any err-named identifier argument to a sink-named callee, including calls whose helper never serialises it (that is why 111 SendErrorResponse rows are listed as CLOSED-BY-HELPER rather than as leaks).
- Producers: service and repository error strings were read only where a row says so (auth_service.go, configuration_service.go, error_reporting_service.go, local_policy.go, localization_service.go, playlist_service.go). Every other LEAK means "text can include internal detail", not a demonstrated leak.
- Not covered: Go modules other than catalog-api and OCU-CUDA-Sidecar (examples/, docs/qa/lib/ctrctl, catalog-api/tests/automation,

tools/ fixtures were not scanned), JavaScript/TypeScript backends, generated code, and HTTP responses produced by third-party middleware. WebSocket (`handlers/websocket_handler.go`) and SSE (`log_management_handler.go`) code was grepped: error values there go to `h.logger` only (UNCONFIRMED beyond that grep).

- HEAD status was produced by matching each review-base row to the HEAD scan by (file, function, sink, kind, expression) and then reading the helper and the changed line for the rows that moved; line numbers shift, so the match ignores them.